

Algebra II with Trigonometry

Chapter 5.2

Mental Math (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4401268_alg-2trig-h-chp-5-2-note-docx.pdf](#)

1. **Ratio Match** In a right triangle, for an acute angle θ , which ratio is opposite over adjacent?
2. **Reciprocal Idea** If $\sin \theta = \frac{3}{5}$, what is $\csc \theta$?
3. **3-4-5 Triangle** In a right triangle with sides 3, 4, 5, if θ is opposite the side of length 3, what is $\cos \theta$?
4. **Special Angle: 30°** What is the exact value of $\tan 30^\circ$?
5. **Special Angle: 45°** What is the exact value of $\sec 45^\circ$?
6. **Radians Check** What is the exact value of $\sin\left(\frac{\pi}{3}\right)$?
7. **Complementary Angles** In a right triangle, if one acute angle is 35° , what is the other acute angle?
8. **Pythagorean Sense** A right triangle has hypotenuse 13 and one leg 5. What is the length of the other leg?
9. **Angle of Elevation** You stand 10 units from a wall and look up to the top at an angle of elevation of 45° . About how tall is the wall?
10. **Choosing the Right Ratio** If you know an angle in a right triangle, the side opposite it, and want the hypotenuse, which trig ratio should you use?